Poker Genius is the best poker training software. Try it for free at: www.Poker-Genius.com

Chapter 26
Doubling Up: Tournaments, Part I

The largest growth area for poker over the past few years has undoubtedly been in the proliferation of tournaments. From the $10,000 World Poker Tour™ events to $1 or freeroll online megatournaments with thousands of players, tournaments have become the most visible and popular form of the game. There are a number of different structures tournaments can take, but they all fundamentally reduce to a poker contest with two key principles:

Tournament chips are irreplaceable - once all a player's tournament chips are gone, the player is eliminated from the tournament.

The first of these principles results in strategic adjustments related to skill - that is, a concept of passing up small edges that put one's whole stack at risk because of presumed situations that will occur later where larger edges will be available. Players who have an edge over the field should be less willing to commit all their chips for a thin edge because if they lose, they cannot replace those chips with additional cash as they could in a ring game. Conversely, players who have negative EV in the tournament should be more willing to gamble all-in on situations with slightly negative chip expectation.

This "skill adjustment," however, has been far overdone by some players; there is a limit to the type of situation that one can legitimately pass up and still actually have such an edge in the future. After all, as the blinds escalate in a tournament, it becomes more and more difficult to find hugely profitable situations. In addition, occasionally opponents will have big hands at the same time as we do. But this skill adjustment is real, and we can seek to quantify it approximately. We model this skill edge as a recursive effect and later in this chapter present a model that attempts to quantify the effect of a skill edge and provide insight into actual situations.

Rebuy tournaments obviously provide a temporary exception to this rule, although all rebuy tournaments eventually reduce to a tournament that satisfies it. We will revisit the topic of taking rebuys later in this chapter.

The prize pool of the tournament is distributed according to a payout structure that is coordinated with the order of bustout and not the raw accumulation of chips.

The second of these principles provides the main mathematically distinguishing characteristic of tournament play in the late stages when compared with cash games. In cash games, seeking to maximize cash EV means seeking to maximize chip EV; chips in cash games are cash. But in tournaments, this is not always the case. Obviously there is a strong correlation between chip position and prize equity; however, the decoupling of what we will call ***chip equity*** (the number of tournament chips a player holds) and ***prize equity*** (the EV of those chips in terms of the prize pool) can create significant adjustments.

A simple example will help to clarify the divergence of chip equity and prize equity in a tournament due to the payout structure.

Consider a 200-player limit holdem tournament of equally skilled players that has been reduced to 19 players. The tables are now going hand-for-hand. 10th-18th place pay $250, and the payouts then escalate to $6,000 for 1st place. The total prize pool is $20,000. There are 300,000 tournament chips in play, and the blinds are 300-600.

Read more poker books for free at: www.PokerBooks.LT

Poker Genius is the best poker training software. Try it for free at: www.Poker-Genius.com

Three players are extremely short-stacked. Player A has just 1 T100 chip remaining, but has the button at his table. Player B has T400 left and is under the gun. Player C has T500 left and is in also under the gun.

Early on in the tournament, the chips that the players hold are worth a share of the prize pool; A's 100 is worth $^1/_{3000}$ of the $20,000 prize pool or about $6.67. But in this situation, his T100 chip has been significantly increased in value. Consider that if he folds his next hand, both of his shortstacked opponents will be all-in for the blind. Assuming that each of them has a 40% chance of winning the hand against (presumably) the strongest hand in the field against them, A has an 84% chance of surviving into the money and getting $250. So even discounting his chances of actually doubling up and building his chip into a stack, his T100 is worth a minimum of $210. This is a far cry from the $6.67 that his chip is worth *a priori*. The tournament payout structure has given his chip additional value. This value is of course taken away from the values of chips in big stacks; recall that collecting all the chips in the tournament is worth only 20% of the prize pool.

This is perhaps a rather extreme example but situations that are of the same character occur in just about every tournament. In these cases, survival is actually worth something - real money. This is generally the case for very small stacks near the "bubble". Their chair is actually worth more than their chips. In addition, at steps in the money, there is incentive for small stacks to play more tightly and attempt to move up, though this is only valid to a point. Some authors have (wrongly) taken this principle and attempted to extend it into the early-game, arguing that strong players should attempt to survive at almost all costs, avoiding confrontations in the early going and attempting to make the money and then making a push toward first.

We believe that this view sacrifices equity because skill-independent chip equity and prize equity are closely tied together in the early going. There simply aren't enough chances to get one's money in as a substantial favorite to justify the loss of tournament chip expectation from the resulting timid, exploitable play. This view also tails because even with the flatter structures now favored by many tournament directors (at this writing 20-25% to first place), a large percentage of the prize pool is concentrated in the top three places and the number of tournament chips held by a player is a strong predictor of that player's chance of finishing in the top three places.

We advocate, then, a strategy which is mindful of the necessary adjustments brought on by the payout structure along with those that a skilled played should make to ensure his repeated ability to apply his skill, but which recognizes that these strategy adjustments are fairly minor while the money is still reasonably far away. From an exploitive standpoint, as well, we seek to capitalize on our opponents' misunderstandings of these concepts, particularly near the money. This results in a lot more "just out of the money" finishes, but it also results in more high finishes and higher expectation overall.

Tournament models are generally most useful in attempting to express the ***mapping*** between chip equity and prize equity in tournaments. If we can map chip equity- to prize equity, then we can return to the familiar idea of maximizing EV from decisions. In this case, however, we do not maximize chip EV, but instead the corresponding cash EV. The first model we consider attempts to quantify the first effect (skill) on tournament decision-making in the early and middle stages of a tournament.

The Theory of Doubling Up

Many players will pass up even large edges in the early or middle stages of tournaments, believing (almost certainly wrongly) that even better edges will present themselves. But if this were actually true, then these players would have gigantic edges over the field in terms of equity.

Read more poker books for free at: www.PokerBooks.LT

Poker Genius is the best poker training software. Try it for free at: www.Poker-Genius.com

Observation does not bear this out. Nonetheless there is an effect - we seek to measure and quantify the effect, and use it as a guide in making accurate decisions at the table.

To assess the value of different tournament stacks, we can use a variety of different models. These models are useful in different stages of the tournament. Some of these models, such as the ones we will introduce in the next section, are primarily useful when the players are near or in the money and the problem is allocating the prize payouts such that each player gets a fair share. The Theory of Doubling Up, by contrast, is intended to be applied in the opening and middle stages of the tournament as a guide to decision-making.

First, let us consider a winner-take-all tournament. In winner-take-all tournaments, excepting skill considerations, a player's equity in the tournament is proportional to his chip stack. We then further make an important assumption: *The chance of doubling a certain player's chip stack before that player busts is constant throughout the tournament.*

There are a number of effects that work on a player's chance of doubling up throughout the tournament. First, there is a mild selection effect. Stronger players tend to survive longer, and the longer a tournament goes, the stronger the overall field will be (on average). This causes all players' chance of doubling up to fall as the tournament progresses.

Secondly, the increasing blinds tend to compress all players' chances of doubling up toward 50% by reducing their latitude to apply their skill (or alternatively, their lack thereof). In the early going, skilled players will have plenty of opportunities to gain chips. Hence, doubling up early is better than doubling up later.

However, neither of these effects is particularly strong, and so we are comfortable taking the above assumption to be true in modeling tournament equities.

Based on this assumption, we can construct a model of a winner-take-all tournament with X players of equal skill who begin with equal stacks. Consider one single player. Let E be his chance of winning the tournament outright, C be his (constant) chance of doubling his stack before busting, and N be the number of times he must double his stack in order to win.

Then we have the simple relationship

$$E = C^N \qquad \textbf{(26.1)}$$

For example, there are 128 players, he must double his stack seven times 1-2-4-8-16-32-64-128. Since each of these doubles is roughly independent, the chance of doubling seven times is C^7.

With this formula, if we have any two of these quantities we can calculate the third. Let's consider the situation at the beginning of the tournament. Since all the players are equally skilled, we know that our player has an equity $E - 1/X$. We also know that in order to win the tournament, our player must increase this stack from its starting value to X times that amount. The number of doubles that this requires can be expressed as:

$$X=2^N$$

So, for a 2-player tournament, N would obviously be 1, as one double would be required to win, while for a 128-player tournament, N is 7. More generally:

$$N = \log_2 X$$

Read more poker books for free at: www.PokerBooks.LT

Poker Genius is the best poker training software. Try it for free at: www.Poker-Genius.com

Substituting:

$1/X = C^{log2X}$

$log_2 1/X = (log_2 X) log_2 C$
$log_2 C = -1$
$C = 0.5$

Of course, this is what we would expect for a winner-take-all tournament where all players are equally skilled; that each player would have a 50% chance of doubling before busting from any stack size. It is easily verified by plugging in other values of A" (representing different stack sizes) that chip values are exactly linear for players of equal skill.

This methodology also generalizes to situations where the players are of unequal skill. For example, consider a hypothetical player A playing in a 100-player tournament where he has EV of 1 buyin net (2 buyins per tournament gross). We can calculate his *C* by plugging in his initial values:

$E = C^N$
$^2/_{100} = C^{\log2\ 100}$
$0.02 = C^{6.643856}$
$C = 0.5550$

So A's chance of doubling his current stack before busting is 55.5%. Now we can calculate his chance of winning the tournament from any stack size S (where *S* is the number of starting stacks he has) by plugging into the equity formula $E = C^N$. *N*, of course, is the number of doubles it will take to reach 100.

So for a stack size of 2 (double the initial stack), A has equity of $E = C^{\log2\ 50}$, or 0.0360. Note that this value is more than the *a priori* value of the chips (which would of course be 0.02), but less than double A's chance of winning the tournament from the start.

What this model does, in effect, is provide us a means for evaluating marginal tournament equity decisions ***including the skill of the player making the decisions.*** The qualitative judgment of "I have an edge over the field, so I should protect my chips" is quantified into an adjusted equity for any stack size that utilizes the assumptions that underlie the model. By doing this, this model provides a mapping of chip equity to tournament equity. To see how the model might work in practice, consider the following marginal situation from a no-limit holdem tournament.

The blinds are 75-150. Player B, with a stack of 3000 (250 players played with starting stacks of 1500), raises to 450 on the button with Qs Ts. The small blind folds and the big blind calls. The flop comes Ks 8c 2s. The blind checks and player B bets 500. The blind check-raises all-in (and has B covered). B estimates his equity' at 36% and is faced with the decision of calling 2050 to win 4025.

An analysis based purely on tournament chip values would indicate that calling is clear - (0.36) (6075) -2050 = 137 chips, for nearly a tenth of a buyin worth of profit. But let's assume that Player B had an edge of $^3/_4$ of a buyin over the field at the beginning of the tournament. Calculating *C* for Player B:

$^{1.75}/_{250} = C^{\log2\ 250}$
$C = 0.5364$

Read more poker books for free at: www.PokerBooks.LT

Poker Genius is the best poker training software. Try it for free at: www.Poker-Genius.com

Now we can calculate the equity *E* of the three scenarios:

If B calls and wins, he will have 6075 chips, or 4.05 starting stacks.

$E = 0.5364^{\log2\,(250/4.05)} = 0.024603$ buyins.

If B calls and loses, he will have 0 chips and have 0 equity.

If B folds, he will have 2050 chips, or 1.36667 starting stacks.

$E = 0.5364^{\log2\,(250/1.36667)} = 0.009269$ buyins.

So B's equity from calling is (0.024603)(0.36), or 0.008857 buyins. Comparing this to B's equity from folding, we find that calling is actually incorrect given B's skill edge over the field.

Another result that can be easily derived from the Theory of Doubling Up includes the "coinflip" problem - what chance of winning do you need to hi order to accept a coinflip for all your chips on the first hand? Since $E_0 = C^N$ at the start of the tournament, if you call and double up, your equity becomes $E_1 = C^{N-1}$. If you take a coinflip with chance of winning W, then your equity becomes WE_1. If you decline, then your equity is E_0. Setting these things equal, we get:

$C^N = WC^{N-1}$, *or* $W=C$.

So you are indifferent to accepting a coinflip if your chance of winning the coinflip is equal to C. Yet consider that many players have a stated preference that they would decline a QQ vs AK (57-43) confrontation in the early tournament. By applying the Theory of Doubling Up, we can find out something about what these players believe their equity in, for example, a 250-player tournament to be.

$E = C^N = (0.57)^8 = 0.01114$

...or about 2.85 buyins per tournament. In order for it to be to correct to decline a 57-43 confrontation with no dead money in the pot, one has to have nearly three times the average equity in the tournament. Our observations lead us to believe that having this win rate in a typical tournament field would place such a player among the best players in the world.

Another interesting effect of the Theory of Doubling Up is what it indicates about players with negative expectation. Of course our readers are all winning players, but nevertheless, the mean result of all players in a tournament is to lose the entry fee, so someone must be losing. The Theory of Doubling Up, then, indicates that losing players should be willing to commit their chips even in marginally bad situations. This is a result of the idea that losing players should encourage variance, as it is their best chance to win. The more that other players get a chance to apply their skill, the worse off the losing player will be. Hence, they should be willing to commit all their chips in zero-EV situations or even slightly bad ones.

The Theory of Doubling Up also has some important implications for multi-way confrontations, ha poker, in general, the "third man in" to an all-in confrontation needs quite a strong hand. This effect is exacerbated by the tournament structure. Consider a situation on the flop where two players are all-in for a pot-sized bet and a player must consider calling all-in for the third bet. In a situation where chips are linear, he would need just a '/4 chance of having the winner to make the call breakeven. In the three-way confrontation, however, the player needs a much stronger hand. If he wins, his chips will be worth four times as much as if he folds; that is, he will have

Read more poker books for free at: www.PokerBooks.LT

Poker Genius is the best poker training software. Try it for free at: www.Poker-Genius.com

reduced his necessary doubles by 2. If his original chance of doubling up was, for example, 55%, he needs 30.25% of the pot just to break even equitywise, compared to 25% chipwise. This is because so often he will be broke and have no future opportunity to apply his skill.

In addition to the form we presented previously, we can show the Theory of Doubling Up formula in a different manner. We have the following from the previous discussion:

$E = C^N$

We can transform this using the rules of logarithms. Suppose we have a stack size S (which is the fraction of total chips in the tournament). Then the number of doubles we need is:

$N = \log_2 (1/S) = -\log_2 S$

$E = C^{-\log 2\, S}$
$\ln E = (-\log_2 S)(\ln C)$
$\ln E = (\ln S)\,(-\log_2 C)$
$\log_S E = -\log_2 C$
$E = S^{-\log 2\, C}$

We denote the quantity -log2 C as β. making this formula:

$$E = S^{\beta} \qquad \textbf{(26.2)}$$

Here we have a restatement of Equation 26.1; only now it is our stack raised to a constant power. We can go further with the ideas of the Theory of Doubling Up and simplify the calculations as well as incorporate ideas that affect our equity estimates. Previously, we have dealt only with winner-take-all tournaments. However, we can additionally incorporate a structured payout into the Theory of Doubling Up.

To understand this methodology, consider a winner-take-all match play tournament with 256 players. Then a single player's equity in the tournament is simply equal to (256) (C^8), or his chance of doubling up seven times (that is, winning matches seven times) multiplied by the number of buyins he wins. We can rewrite this expression as:

$(256)(C^8) = (2C)^8$

In fact, in general, we can modify' the $E = C^N$ equation to express equity in the total prize pool of X buyins as follows:

$E_X = XC^N$
$E = (2C)^N$

(since N is the number of doubles required)

Now consider the same tournament, except that the total prize pool is divided between first and second place equally - that is, 1st and 2nd are both 128 buyins. His equity here is actually $(128)(C^7)$, or $(2C)^7$.

How does this relate to the previous equity?

$(2C)^8 > (2C)^7$

Read more poker books for free at: www.PokerBooks.LT

Poker Genius is the best poker training software. Try it for free at: www.Poker-Genius.com

$C > 1/2$

By splitting the first prize equally among the first two players, our player's equity is reduced if his chance of doubling is greater than $1/2$, while it is increased otherwise. This makes sense - basically, what's happening here is that instead of getting his chance of winning the last match times 256, he gets 128 instead. So his chance of winning the last match is simply replaced by $1/2$.

What this reflects is that in tournaments where not all the money is paid to first place, the opportunity to apply skill is reduced, because a player who acquires all the chips only acquires a fraction of the prize pool. So in the second case discussed above (where the prize pool is split between first and second place), our skillful player has less equity by one factor of *2C*. According to our equity estimates from the Theory of Doubling Up (which of course applies directly and not as a model in this case), the skill multiplier for this tournament is the same as if the tournament contained 128 players instead of 256.

We call this the ***effective tournament size*** - essentially what we're doing is quantifying the degree to which the structured payouts reduce the opportunity to apply skill. We can say that as a close approximation, a tournament of 256 players where the first and second prizes are equal has the same latitude for skill as a 128-player tournament where it's winner-take-all. We denote the effective tournament size (generally) as *k*, and define *k* more formally to be a value such that the following equation is satisfied for a given prize distribution:

$$(2C)^{\log_2 k} = E_N \qquad \textbf{(26.3)}$$

where C is the chance of doubling up and E_N is the number of buyins the player wins in the tournament with the prize structure intact. Then k will be the size of the equivalent tournament (for the Theory of Doubling Up) with a winner-take-all structure.

Of course, tournaments don't often split the prize pool equally between first and second place, preferring to instead give a larger piece of the pool to first. Suppose that our hypothetical match play tournament instead has payouts of f of the prize pool to first place and $(1 - f)$ to second. Now our player's equity becomes:

(total prize pool)[(1st money)(p(1st)) + (2nd money)(p(2nd))]
$(256)(f\,(C^8) + (1 - f)(C^7 - C^8)$
$(256)C^7\,(2fC + 1 - C - f)$

We know that for our hypothetical whining player, C will be a little bit higher than 0.5; for example 0.53. We'll use the symbol 5 to denote the difference between C and 0.5,

$\delta = C - 0.5$

Then our equity formula simplifies to:

$(256)\ C^7\ (\delta\ (2f - 1) + 1/2)$

This formula is fundamentally made of three parts:

- $(256)\ C^7$ is our hypothetical player's chance of making it to the finals times the entire prize pool.
- $\delta\ (2f - 1)$ is the "extra" equity he wins because of skill.
- $1/2$ is his base share of the prize pool.

Read more poker books for free at: www.PokerBooks.LT

Poker Genius is the best poker training software. Try it for free at: www.Poker-Genius.com

Using this equity formula, we can find the effective tournament size for this tournament.

Let d equal $\log_2 k$, the number of doubles that would be required in a winner-take-all tournament of size k.

There exists some winner-take-all tournament such that:

$(2C)^d = E_N$

In this case, E_N is equal to our equity expression, so:

$(2C)^d = (256)C^7 (\delta (2f - 1) + 1/2)$
$(2C)^{d-7} = 2 (\delta (2f - 1) + 1/2)$
$(d - 7)(\ln 2C) = \ln (2 (\delta (2f - 1) + 1/2))$

$(d - 7) = \ln (2 (\delta (2f - 1) + 1/2))/ \ln (2C)$
$d = \ln (2 (\delta (2f - 1) + 1/2))/ \ln (2C) + 7$
$k = 2^d$

Suppose that f *is* 154/256 and C is 0.52. Then

$$d = \frac{\ln (2(0.02(2\left(\frac{154}{256} - 1\right) + 0.5)}{ln2C} + 7 \approx 7.206$$
$k = 2^d = 147.68$

Notice that k here is dependent on the value of C that we use. However, it turns out that the value of k is relatively insensitive to the chosen value of C as long as C falls within the usual parameters. We will utilize this fact in extending this discussion to payout structures with more paid places.

So our effective tournament field size here is actually about a hundred players smaller than the actual field size, simply as a result of the prize structure being divided. Note that this effective field size is used as the field size in the Theory of Doubling Up equations. Also, the E term must be multiplied by this number of buyins to find the proper equity.

For example, to find the equity of a player with C = 0.53 who has won his first four matches (thus needing 4 more doubles to win), we can use the following:

$kE = C^{log2\ (k/S)}$
$(147.68)(E) = (0.53)^{\log2\ (147.68/16)}$
E = 19.287 buyins

We can also calculate this directly as a check:

Chance of making the finals: $0.53^3 = 0.14888$

Chance of winning in the finals: 0.53

(0.14888)[(0.53)(154) + (0.47)(102)] = 19.288 buyins

Read more poker books for free at: www.PokerBooks.LT

Poker Genius is the best poker training software. Try it for free at: www.Poker-Genius.com

Most tournaments are not match play, and most are not limited to two places paying. However, accepting some assumptions about the likelihood of brushing in different places, we can actually generalize this result to find the effective tournament field size (and by doing so, be able to generate the proper skill adjustments) for any payout structure.

Assumption:

The chance of a player's finishing in place N is equal to his chance of doubling up to a stack size of 1/N of the chips in play, less his chance of finishing in a higher place.

So a player's chance of finishing first is his chance of doubling up to all of the chips in the tournament. His chance of finishing second, then, is his chance of doubling up to half the chips in the tournament less his chance of finishing first. His chance of finishing third is his chance of doubling up to a third of the chips in the tournament, less this chance of finishing first or second, and so on. This series does hi fact sum to one: even though it's true that it's not required that a player double up to 1/N chips in order to finish Nth, this approximation is a reasonable one.

We can manipulate the Theory of Doubling Up and this assumption to find an effective tournament field size for any tournament. The following discussion is a little challenging, so we mark it off and you can feel free to skip it

A player's equity (in buyins) is given by the following formula:

$$E_N = \sum_{i=1}^{x} p_i v_i$$

where pi is the probability of placing in i^{th} place, v_i is the prize payout for finishing in i^{th} place, and *X is* the total number of players in the tournament.

The probability of finishing in each place is equal to the probability of doubling up to the appropriate chips minus the chance of doubling to the appropriate chips for the next higher spot:

$$p_i = C^{\log_2(\frac{X}{i})} - C^{\log_2(\frac{X}{i-1})}$$

We define $N = \log_2 X$. This is the number of doubles it would take if this were a winner-take-all tournament. Substituting into the equation for the probabilities of finishing in each place:

$$p_i = C^{N - \log_2 i} - C^{N - \log_2 (i-1)}$$

This last equation only holds true if $i > 1$; at $i = 1$, $p_i = C^N$.

We now define a separate series of terms, related *to* p_i.

$$q_i = \sum_{i=1}^{X} p_i$$

So q_i is the "cumulative" probability that we will finish in the top i players; q_i is the chance that we will finish first, q_2 is the chance that we will finish first or second, and q_N is the chance that we will finish in N^{th} place or higher.

Read more poker books for free at: www.PokerBooks.LT

Poker Genius is the best poker training software. Try it for free at: www.Poker-Genius.com

Plugging in pi values, we have:

$q_i = C^{N - log2\ i}$

Now consider the prize structure of our tournament. Calling the total prize pool P, where v_i, is the payout for finishing i^{th} we have:

$$P = \sum_{i=1}^{X} v_i$$

Then w_i is the incremental increase in guaranteed money for finishing in i^{th} place instead of $[i + 1)^{th}$ place such that:

$$w_i = \sum_{i=1}^{X} v_i - v_{i+1}$$

with the special rule that $w_X = v_X$

We also have the following identity:

$$P = \sum_{i=1}^{X} iw_i$$

because

$$\sum(iw_i) = \sum(i(v_{i-v} - v_{i+1}))$$

$$= \sum(iv_i) - \sum\left(iv_{(i+1)}\right)$$
$$= \sum(iv_i) - \sum\left((i-1)v_i\right)$$
$$= \sum(v_i) = P$$

The total prize pool is equal to the money guaranteed for finishing i^{th} or higher times the number of players who finish there, summed for all i values.

We saw previously that

$$E_N = \sum_{i=1}^{x} p_i\, v_i$$

A player's equity is equal to the value of each finish times the probability of that finish, summed over all finishes.

However, we can also say the following:

$$E_N = \sum_{i=1}^{x} q_i\, w_i$$

A player's equity is equal to his chance of finishing in each place or higher multiplied by the

Read more poker books for free at: www.PokerBooks.LT

Poker Genius is the best poker training software. Try it for free at: www.Poker-Genius.com

incremental money he is guaranteed by getting to that spot.

To show this, suppose that we have just three payouts (this approach generalizes to more payouts), v_1, v_2, and v_3 and the probabilities of finishing each of these spots are p_1 ,p_2, and p_3, respectively.

We then have:

$q_1 = p_1$
$q_2 = p_1 + p_2$
$q_3 = p_1 + p_2 + p_3$

and

$g_1 = v_1 - v_2$
$g_2 = v_2 - v_3$
$g_3 = v_3$

We can now show that $q_1g_1 + q_2g_2 + q_3g_3$ is equivalent to our equity:

$q_1g_1 + q_2g_2 + q_3g_3$
$p_1(v_1 - v_2) + (p_1 + p_2)(v_2 - v_3) + (p_1 + p_2 + p_3)(v_3)$
$p_1v_1 + p_2v_2 + p_3v_3$

With this, we have our equity for the game. What we are looking for is the k value such that the following equation is satisfied:

$$2N \sum_{i=1}^{x} q_i\, w_i = (2C)^D$$

where D is the number of doubles necessary to reach k. players, or $\log_2 k$. This presumes that w_i is denominated in buyins and not fractions of the prize pool.

$$2N \sum_{i=1}^{x} C^{N-\log_2 i} w_i = (2C)^D$$

$$\sum_{i=1}^{x} (2C)^N C^{-\log_2 i} w_i = (2C)^D$$

$$\sum_{i=1}^{x} C^{-\log_2 i} w_i = (2C)^{D-N}$$

$$\sum_{i=1}^{x} i(2C)^{-\log_2 i} w_i = (2C)^{D-N}$$

Next we make use of the idea that k is insensitive to the value of C chosen. Suppose we choose C very close to 0.5. In keeping with convention, we use ε to denote twice the distance between 0.5 and C (The value 0.5 exactly is a discontinuity because it involves dividing by zero).

Read more poker books for free at: www.PokerBooks.LT

Poker Genius is the best poker training software. Try it for free at: www.Poker-Genius.com

$$\sum_{i=1}^{x} i(1+\varepsilon)^{-\log_2 i} w_i = (2C)^{D-N}$$

Assuming e is close enough to 0.5, we can approximate the exponential of expansion as follows:

$(1 + \varepsilon)^x = 1 + x\varepsilon$.

$$\sum_{i=1}^{x} i w_i (1 - (\log_2 i)\varepsilon) = (1 - (N - D)\varepsilon)$$

The iw_i term simply sums to 1, and we are left with:

$$\sum_{i=1}^{x} i w_i (\log_2 i) = (N - D)$$

N - *D* can be thought of as the number of doubles that are "missing" from the tournament when the prize structure is incorporated. To find the actual effective tournament field size, we can divide X by 2^{N-D}

This much simpler formula can be used to find the effective prize structure for any tournament. We can test it on our previously examined tournament.

$g_1 = {}^{154}/_{256}$
$g_2 = {}^{102}/_{256}$

${}^{204}/_{256} = N - D$

$k = 256 / 2^{204/256} = 147.35$ where $D = \log_2 k$

The slight variance here is caused by the fact that when we first looked at this tournament, we chose a *C* of 0.52.

Key Concepts

- The Theory of Doubling Up allows us to evaluate equity decisions in the early to middle portions of a tournament by using a constant "chance of doubling up" and finding the number of doubles required to get all the chips. Using these two measures, we can effectively model tournament skill and get a sense for the marginal values of chips.
- The $E=C^N$ form of the Theory of Doubling Up can be further refined to understand the effect that a flat payout structure has on skill. When money is given to a place other than 1st, the effect of skill is diminished.
- We can solve for an effective tournament size (see Equation 26.3), which is the size of a winner-take-all tournament where the skill adjustments are the same as our structured payout tournament.

Read more poker books for free at: www.PokerBooks.LT